climatekit: Unified Climate Indices for Temperature, Precipitation, and Drought in R

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Abstract The **climatekit** package computes the standard suite of climate indices from daily weather observations in a single R interface. Thirty functions cover temperature (frost days, ice days, summer days, growing and heating degree days, tropical nights, diurnal range, growing-season length, warm-spell duration), precipitation (wet and dry spells, heavy-precipitation days, one-day and five-day maxima, precipitation intensity), drought (Standardised Precipitation Index, Standardised Precipitation-Evapotranspiration Index, potential evapotranspiration via the Hargreaves method), agroclimatology (Huglin, Winkler, Branas, frost dates), and thermal comfort (wind chill, heat index, humidex, fire danger). Definitions follow the Expert Team on Sector-specific Climate Indices recommendations (Zhang et al. 2011) where applicable. Every function returns a tidy data frame with a period, a value, an index name, and a unit, so outputs compose directly. The package is pure computation, has no API calls, and is available on CRAN.

1 Introduction

The **climatekit** package consolidates thirty climate indices behind a single consistent interface. Each function takes daily vectors (temperature, precipitation, humidity) with an accompanying date vector and returns a data frame with four columns: `period` (the reporting period, typically annual or monthly), `value` (the index value), `index` (a character name), and `unit`. Users can feed any output directly into plotting, tabulation, or further aggregation without reshaping.

Climate-index analysis is a routine operation across applied climate work: national meteorological services publish annual index tables, regional climate assessments benchmark stations against one another, agricultural and viticultural research tracks growing-season indices, insurance underwriters use drought indices in parametric products, and public-health authorities monitor heat-index exceedances. Until now, the R ecosystem has exposed parts of this analysis piecemeal, with separate packages for drought indices (**SPEI**), for standard climate-change indices (**climdex.pcic**, archived), and for various agricultural indices. **climatekit** unifies the set.

The package has no network access, no external data dependency, and no runtime requirements beyond **cli** and base R. Users pair it with a data-acquisition package (**readnoaa** for NOAA's Global Historical Climatology Network, **rnoaa** for broader NOAA holdings, or direct downloads from a national met service) and pass the resulting daily series straight in.

2 Background

Climate-index standardisation dates to Frich et al. (2002), who proposed twenty-seven indices targeted at climate-change detection, and continued through the joint CCl/WCRP/JCOMM Expert Team on Climate Change Detection and Indices (ETCCDI), now the Expert Team on Sector-specific Climate Indices (ET-SCI). Zhang et al. (2011) synthesised the ET-SCI core set, which informs most modern national climate assessments. Applied analyses of the ETCCDI indices on observed station records (Alexander et al., 2006) and on gridded reconstructions (Donat et al., 2013) document robust multi-decadal trends. Sillmann et al. (2013) evaluate the same indices across CMIP5 climate-model output, establishing the index suite as the bridge between observational and model-based climate-change attribution.

Temperature indices. The core definitions are threshold counts (frost days where $T_{\min} < 0°\mathrm{C}$; tropical nights where $T_{\min} \geq 20$; summer days where $T_{\max} \geq 25$; ice days where $T_{\max} < 0$) and degree-day accumulations (growing degree days above a base, heating degree days below a reference, cooling degree days above a reference). Diurnal range, growing-season length, and warm-spell duration round out the temperature family.

Precipitation indices. Consecutive dry and wet days summarise spell length. Total accumulation, heavy-precipitation exceedances, and one- and five-day maxima summarise intensity. The simple daily intensity index (SDII) is the average daily total on wet days. The ETCCDI-standard thresholds for heavy-precipitation counts are the 95th (R95p) and 99th (R99p) percentiles of the wet-day distribution

at each station; `ck_heavy_precip()` accepts an explicit `threshold` argument so users can supply either a fixed value or a station-specific percentile computed from the data.

Drought indices. McKee et al. (1993) introduced the Standardised Precipitation Index (SPI), which fits a gamma distribution to precipitation accumulated over a chosen window (three, six, twelve months) and standardises to a standard normal. Values below minus one indicate moderate drought; below minus two, extreme drought. Vicente-Serrano et al. (2010) extended this to the Standardised Precipitation-Evapotranspiration Index (SPEI), which incorporates potential evapotranspiration via a climatic water balance. Stagge et al. (2015) compare candidate distributions for both indices and find that the gamma family (for SPI) and the generalised logistic (for SPEI) perform best in terms of normality-of-residuals tests; **climatekit** uses these defaults. Potential evapotranspiration is computed by the Hargreaves and Samani (1985) method from temperature extremes and extra-terrestrial radiation.

Agroclimatic indices. The Huglin heliothermal index (Huglin, 1978) accumulates heat summed between a base temperature and a latitude-adjusted day length; it is the primary viticultural suitability index in temperate Europe. Tonietto and Carbonneau (2004) propose a multicriteria climatic classification that combines the Huglin index with cool-night and dryness indices, and supply the classification bands (HI-3 for cool varieties, HI-2 for temperate, HI-1 for warm, plus subdivisions) used in viticultural planning. The Winkler index (Winkler et al., 1974) is a similar cumulative heat metric used widely in California. Branas, Bernon, and Levadoux's index is the French viticultural analogue. First-frost and last-frost dates set the frost-free growing window.

Thermal comfort. The US National Weather Service's heat index (Rothfusz, 1990) converts temperature and humidity into an apparent-temperature equivalent; Canadian humidex (Masterton and Richardson, 1979) does the same with a different functional form. Wind chill (Osczevski and Bluestein, 2005) is the cold-weather counterpart. Fire-weather indices combine temperature, humidity, wind, and precipitation history.

3 Package design

3.1 Architecture

climatekit is pure R with no compiled code. Runtime imports are **cli**, `stats`, and `tools`. R 4.1.0 or later is required. The test suite contains over a hundred tests covering every exported function and every index definition.

3.2 Uniform output format

Every index function returns a data frame with four columns: `period`, `value`, `index`, and `unit`. The `period` is a `Date` keyed to the start of the reporting window (year or month); the `value` is numeric; `index` and `unit` are character strings. This invariant means output from any function composes directly into a larger panel via `rbind()`, and plotting code is shared across index families.

3.3 Uniform input signature

Every index function takes one or two daily numeric vectors plus a `dates` vector of matching length. Where appropriate, a `period` argument selects annual or monthly aggregation. Users do not pass data frames; users pass the three or four vectors they have.

3.4 Generic dispatcher and metadata

`ck_compute()` dispatches to any index by name, letting users select the index programmatically from a configuration file or argument. `ck_available()` lists the indices that can be computed given a supplied set of inputs; `ck_metadata()` returns each index's definition, unit, reference, and typical use.

3.5 Reproducibility

Numerical results are deterministic given the input. The SPI and SPEI implementations use gamma-distribution fitting, which is deterministic up to the solver's numerical tolerance. All index definitions are documented with an explicit reference to the underlying paper in the function's `@details` block.

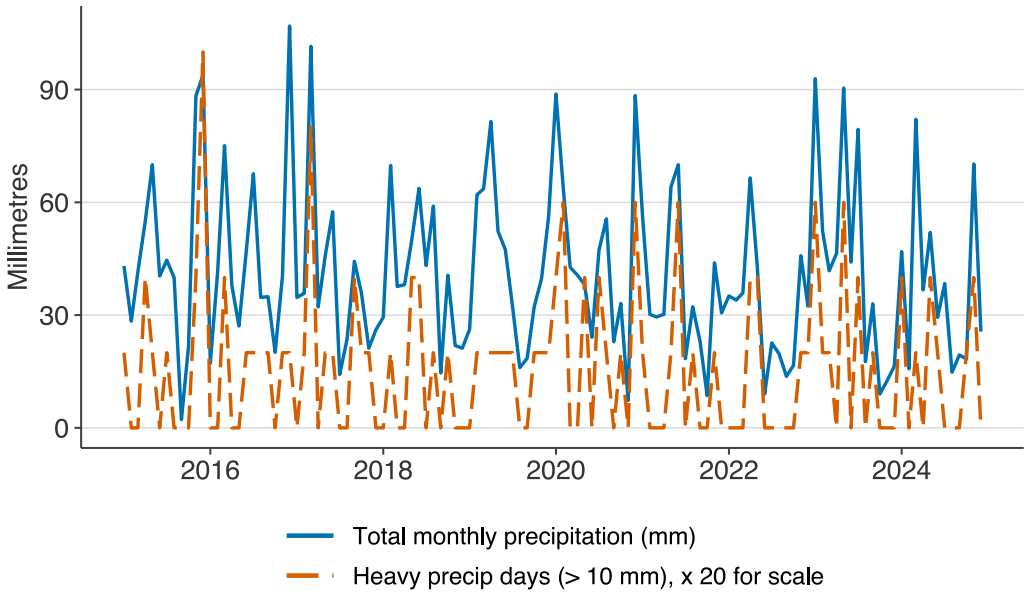

Figure 1: Monthly precipitation indices on a 10-year synthetic temperate site. Blue: total monthly precipitation in millimetres. Red dashed: count of heavy-precipitation days (threshold 10 mm) scaled up by a factor of 20 for visual comparability. Total precipitation shows seasonal variation; heavy events are concentrated in wetter months.

4 Temperature indices

Four threshold-exceedance counters cover the core temperature indices. Cold extremes are counted by `ck_frost_days()` (minimum below 0°C) and `ck_ice_days()` (daytime maximum below zero). Hot extremes are counted by `ck_summer_days()` (maximum at or above 25) and `ck_tropical_nights()` (minimum at or above 20).

Three degree-day accumulators report cumulative heat above or below a reference temperature. Agricultural work uses `ck_growing_degree_days()` to sum degree-days above a user-specified base. Energy-demand applications use the heating and cooling counterparts, `ck_heating_degree_days()` and `ck_cooling_degree_days()`.

`ck_diurnal_range()` reports the mean daily temperature range. `ck_growing_season()` returns the frost-free growing season length. `ck_warm_spell()` counts days in spells exceeding a high-percentile threshold. Figure 5 in the case study below shows these indices applied to seventy-five years of real NOAA data from the New York Central Park station.

5 Precipitation and drought indices

`ck_total_precip()`, `ck_dry_days()`, and `ck_wet_days()` summarise accumulation and spell counts. `ck_heavy_precip()` and `ck_very_heavy_precip()` count exceedances of user-specified thresholds. `ck_max_1day_precip()` and `ck_max_5day_precip()` return rolling maxima. `ck_precip_intensity()` returns the simple daily intensity index.

Figure 1 shows two precipitation indices on the same synthetic site: total monthly precipitation (with substantial seasonal variation) and the count of heavy-precipitation days (greater than 10 mm), with the latter scaled up by a factor of 20 for visual comparability.

`ck_spi()` returns the Standardised Precipitation Index at a user-specified accumulation scale (most commonly 3, 6, or 12 months). Figure 2 shows SPI-3 and SPI-12 on the same synthetic site. Short-scale SPI captures short-term wet and dry spells; long-scale SPI smooths out noise and reveals sustained anomalies. Dotted horizontal lines mark conventional moderate-drought and moderate-wet thresholds (± 1) and severe thresholds (± 1.5).

`ck_spei()` computes the Standardised Precipitation-Evapotranspiration Index, using `ck_pet()` for potential evapotranspiration via the Hargreaves and Samani (1985) method.

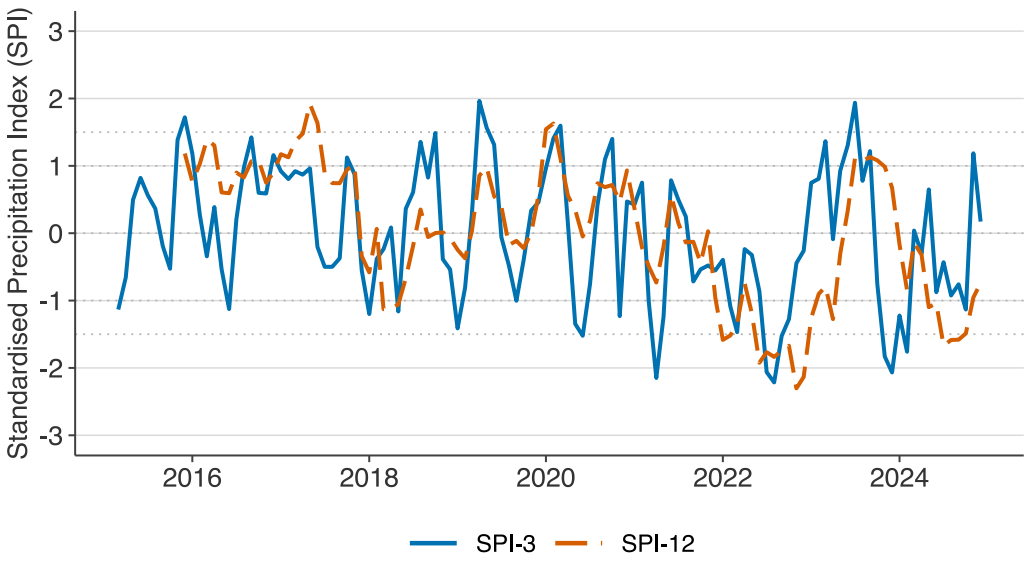

Figure 2: Standardised Precipitation Index at 3-month and 12-month scales on a 10-year synthetic site. Blue solid: SPI-3. Red dashed: SPI-12. Dotted horizontal lines at ±1 and ±1.5 mark moderate and severe drought (negative) and wet (positive) thresholds. The 12-month index smooths short-term variability; the 3-month index captures individual wet and dry spells.

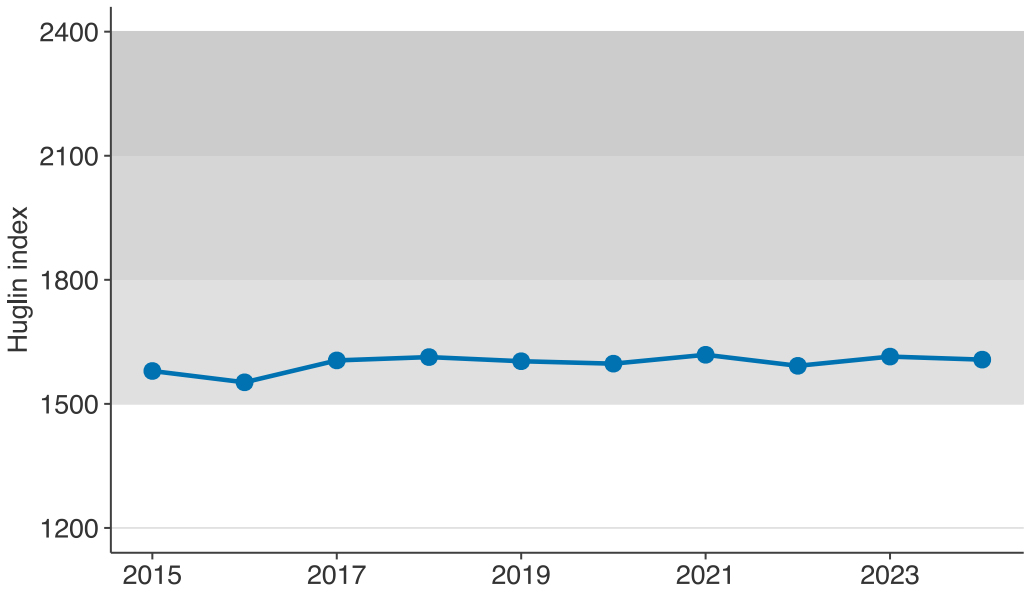

Figure 3: Huglin heliothermal index on a 10-year synthetic temperate site at latitude 51.5°N. Shaded horizontal bands mark approximate viticultural suitability ranges: below 1500 is too cool, 1500 to 1800 is Cool (HI-3, suitable for early-ripening varieties), 1800 to 2100 is Temperate, 2100 to 2400 is Warm, above 2400 is Hot. The sample falls mostly in the Cool range.

6 Agroclimatic indices

`ck_huglin()` returns the Huglin heliothermal index, accumulated from April through September (Northern Hemisphere) from daily maximum and mean temperatures weighted by a latitude correction. `ck_winkler()` returns the Winkler index (a simpler cumulative growing-degree-day measure used in California viticulture). `ck_branas()` returns the Branas product index. `ck_first_frost()` and `ck_last_frost()` report frost dates each year.

Figure 3 shows the Huglin index on the synthetic temperate site. Grey horizontal bands mark approximate viticultural classification ranges; the site plots within the Cool to Temperate range in most years of the sample.

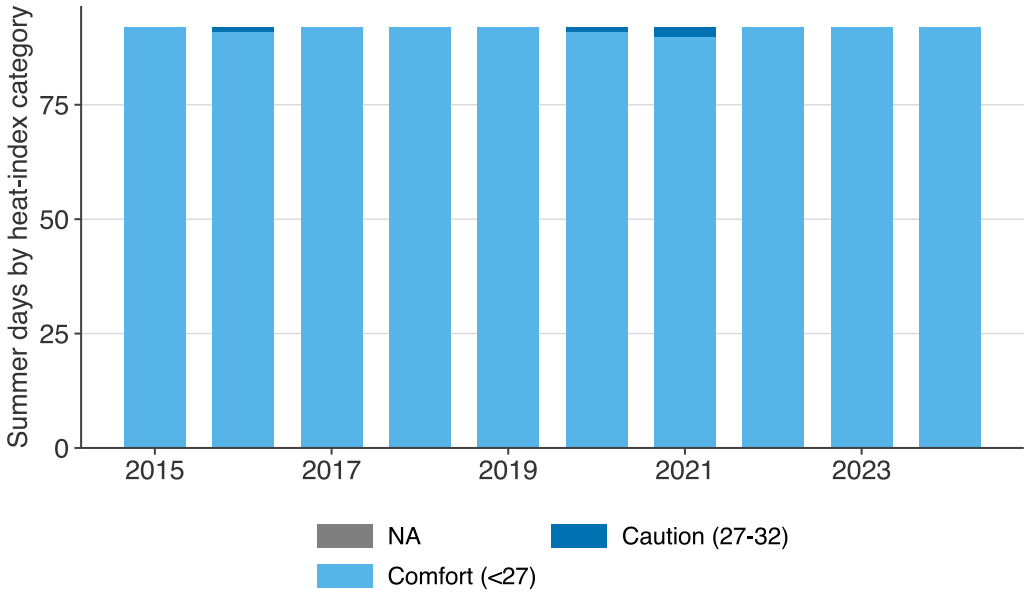

Figure 4: Summer (June-August) heat-index distribution across 10 years of a temperate synthetic site, bucketed by US NWS caution categories. Most summer days fall in the comfort (heat index $< 27°$C) or caution (27 to 32) categories; extreme caution days are uncommon; danger and extreme danger are absent in this temperate sample, as expected.

7 Comfort indices

Four comfort functions cover hot and cold extremes. `ck_heat_index()` returns the US National Weather Service heat index (apparent temperature) from mean temperature and relative humidity. `ck_humidex()` returns the Canadian humidex from the same inputs. `ck_wind_chill()` returns wind chill from temperature and wind speed. `ck_fire_danger()` returns a linearised approximation of the Canadian Fire Weather Index (CFFWIS) combining temperature, relative humidity, wind speed, and precipitation. The implementation is intentionally a simplified surrogate for rapid daily computation; users requiring the full iterative CFFWIS should pair with the specialised **cffdrs** package.

Figure 4 shows the distribution of summer heat-index values, bucketed into US National Weather Service caution categories, across ten years of the synthetic site. Most summer days fall in the comfort or caution categories; higher categories are rare at this latitude, as expected.

8 Case study: 75 years of Central Park temperature indices

Figure 5 applies **climatekit** to real NOAA Global Historical Climatology Network Daily (GHCND) data from the New York Central Park station (USW00094728) from 1950 to 2024, a seventy-five-year daily record with zero missing observations in $T_{\min}$, $T_{\max}$, and precipitation. Three temperature indices are computed directly from the raw daily vectors: frost days, summer days, and growing degree days at base 10. A linear trend fitted over the sample returns a decline of 2.6 frost days per decade, a rise of 28 growing-degree-days per decade, and a roughly stable summer-days count. The signs and magnitudes are consistent with the observed warming trends for the US Northeast documented in Alexander et al. (2006) and Donat et al. (2013). The full computation is three function calls on three vectors.

9 Replication

Table 1 lists the thirty indices grouped into five families. The representative function names give the flavour of the interface.

The canonical workflow is four lines:

```
library(climatekit)
frost_days <- ck_frost_days(tmin, dates, period = "annual")
spi3       <- ck_spi(precip, dates, scale = 3)
```

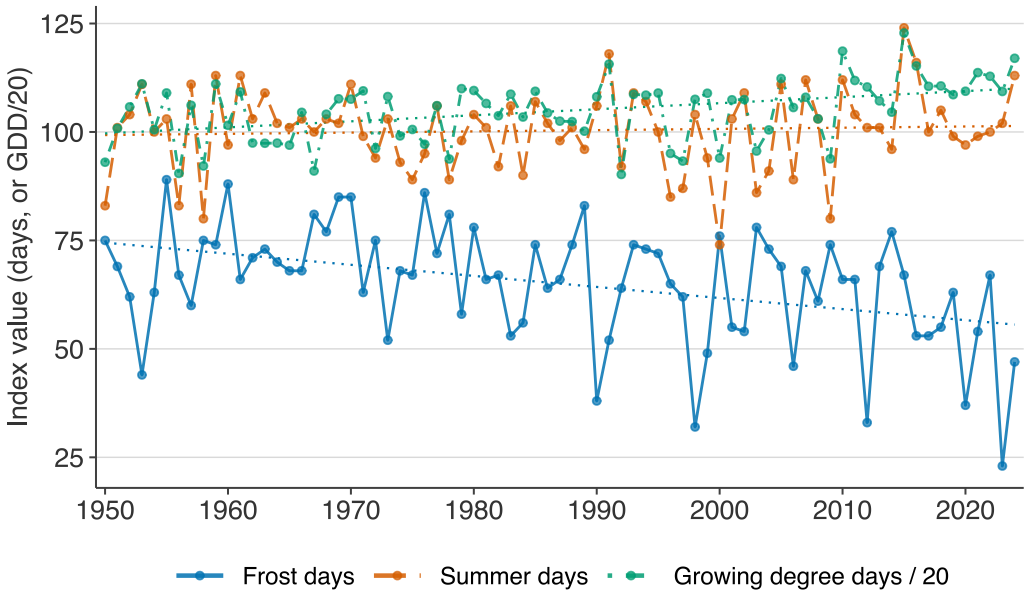

Figure 5: Three temperature indices for New York Central Park (GHCND station USW00094728), 1950 to 2024. Annual counts from raw daily T_{min}, T_{max}, and T_{mean}. Blue solid: frost days ($T_{\text{min}} < 0°\text{C}$), linear trend -2.6 days per decade. Red dashed: summer days ($T_{\text{max}} \geq 25°\text{C}$), trend $+0.3$ days per decade. Green dotdash: growing degree days above base 10°C, divided by 20 for visual scale, trend $+28$ GDD per decade. Dotted lines are OLS trend fits.

Table 1: Index families in climatekit, with count and representative function names. All functions share the `ck_` prefix and return a data frame with columns `period`, `value`, `index`, `unit`.

Index family	Count	Representative functions
Temperature	10	`ck_frost_days, ck_summer_days, ck_growing_degree_d. . .`
Precipitation	8	`ck_total_precip, ck_dry_days, ck_wet_days, ck_heav. . .`
Drought	3	`ck_spi, ck_spei, ck_pet`
Agroclimatic	5	`ck_huglin, ck_winkler, ck_branas, ck_first_frost, . . .`
Comfort	4	`ck_wind_chill, ck_heat_index, ck_humidex, ck_fire_. . .`
Total	30	

```
huglin     <- ck_huglin(tmin, tmax, dates, lat = 51.5)
```

Each call returns a tidy data frame that can be bound directly into a panel with other indices or plotted without reshaping.

10 Limitations

Five limitations apply.

1. **climatekit** computes indices from daily observations. Sub-daily (hourly) indices, including heat-wave indices based on night-time minimum temperatures and apparent-temperature exceedances, are not implemented.
2. Gridded-climate input is not natively supported. Users must reduce a gridded product to a per-cell time series before passing it in. For CMIP6 and similar model output, **terra**-based preprocessing is the natural partner.
3. The SPI and SPEI implementations use the gamma distribution as the parametric family. Users studying very arid regions may prefer the Pearson III family favoured by Vicente-Serrano et al. (2010); this is not currently exposed.
4. Quality control is the user's responsibility. Missing-value handling propagates `NA` through the aggregation functions; users should screen input data for known stations-of-concern flags before computing indices.

5. Station homogenisation is not implemented. Detecting inhomogeneities (relocation, instrument change) in a daily record requires specialised techniques that are out of scope.

11 Appendix: index definitions

Selected formulas used by **climatekit**. Daily observations are indexed by t; aggregations run over a reporting window $\mathcal{T}$ (year, month, or season).

Frost days. $\mathrm{FD} = \sum_{t \in \mathcal{T}} \mathbf{1}\{T_{\mathrm{min},t} < 0^\circ\mathrm{C}\}$.

Growing degree days (base b). $\mathrm{GDD}_b = \sum_{t \in \mathcal{T}} \max(T_{\mathrm{mean},t} - b, 0)$.

Huglin heliothermal index. For latitude ϕ and northern-hemisphere summation over 1 April to 30 September,

$$\mathrm{HI} = K(\phi) \sum_{t=\mathrm{Apr}}^{\mathrm{Sep}} \frac{(T_{\mathrm{mean},t} - 10) + (T_{\mathrm{max},t} - 10)}{2},$$

where $K(\phi)$ is Huglin's day-length correction ranging from 1.02 at 40°N to 1.06 at 50°N.

Standardised Precipitation Index. For precipitation P_s summed over an s-month window, fit a two-parameter gamma distribution to the historical distribution of P_s with shape α and scale β, compute the non-exceedance probability $F(P_s; \alpha, \beta)$, and transform to a standard normal: $\mathrm{SPI}_s = \Phi^{-1}(F(P_s; \alpha, \beta))$. Values below -1 indicate moderate drought; below -2, extreme drought. The gamma-family default follows Stagge et al. (2015).

Heat index (US NWS, Rothfusz form). For air temperature T in °F and relative humidity R in per cent,

$$\mathrm{HI} = c_1 + c_2 T + c_3 R + c_4 TR + c_5 T^2 + c_6 R^2 + c_7 T^2 R + c_8 T R^2 + c_9 T^2 R^2,$$

with the nine Rothfusz coefficients (Rothfusz 1990). The package converts °C inputs internally.

12 Conclusion

Climate-index computation is a daily activity in meteorological services, climate research, agricultural consultancies, and insurance underwriting, and the R ecosystem had not had a single unified implementation. **climatekit** consolidates thirty indices across temperature, precipitation, drought, agroclimatology, and thermal comfort behind a tidy-data-frame output format. The package is pure R, has no data dependencies, and returns output that composes directly into downstream analysis. Planned additions include hourly heat-wave indices, gridded-input helpers, and additional fire-weather variants. Contributions and bug reports are welcome at `https://github.com/charlescoverdale/climatekit`.

Acknowledgements

The CCl/WCRP/JCOMM Expert Team on Climate Change Detection and Indices (ETCCDI) and its successor the Expert Team on Sector-specific Climate Indices (ET-SCI) developed and documented the index definitions on which this package is built. The National Oceanic and Atmospheric Administration publishes the Global Historical Climatology Network Daily (GHCND) dataset that is the natural input source for these indices.

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